

An Introduction to Apache Hive CI

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July 23, 2025



About me

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PMC member of Apache Calcite & Apache Hive

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Services



Cluster



Jenkins



Jenkins

Search (CTRL+K)log in

Dashboard >

Build History

Project Relationship

Check File Fingerprint

Open Blue Ocean

Build Queue

No builds in the queue.

Build Executor Status

Built-in Node

1 Idle

2 Idle

hive-precommit-master-2615-2zv55-xkl1ctc5zkh

(launched...) (suspended)

hive-precommit-master-2615-4t009-2cmv4-25k01

(launched...) (suspended)

hive-precommit-master-2615-542zs-7dx63-m5icd

(launched...) (suspended)

hive-precommit-master-2615-5rsh3-2ccpn-8gm7x

(launched...) (suspended)

hive-precommit-master-2615-5s729-ljbrn-nwrrg

(launched...) (suspended)

hive-precommit-master-2615-77fj9-fxx9d-

(launched...) (suspended)

Welcome to Jenkins!

Log in now to view or create jobs.

Log in to Jenkins →



- General anonymous access is disabled for security reasons
- Authorization requires a GitHub account
- Extra privileges can be granted to Hive committers/PMC per request

Matrix-based security

[illegible]



Jenkins - Available Jobs

<https://ci.hive.apache.org/>

S	W	Name ↓	Last Success
✓	☀	chk	3 hr 54 min #170296
!	☀	hive-flaky-check	24 days #891
✗	☁	hive-nightly	1 mo 22 days #744
✓	☀	hive-nightly-branch-3	11 mo #253
📄	☁	hive-precommit	5 hr 16 min log
✓	☀	space-check	10 hr #1500



Jenkins - Precommit job

<https://ci.hive.apache.org/job/hive-precommit/>

Mirror of Apache Hive

Branches (15)

Pull Requests (231)

S	W	Name	Last Success	Last Failure	Last Duration	F
⊗	☀	branch-2	4 yr 10 mo #4	N/A	6 hr 18 min	☆
⊗	☀	branch-2.3	1 yr 12 mo #33	4 yr 2 mo #18	1 hr 37 min	☆
ⓘ	☀	branch-3	1 yr 3 mo #129	2 yr 5 mo #25	2 hr 9 min	☆
ⓘ	☀	branch-3.1	1 yr 2 mo #27	3 yr 11 mo #15	1 hr 50 min	☆
✅	☀	branch-4.0	9 mo 28 days #30	1 yr 3 mo #17	2 hr 23 min	☆
⊗	☀	branch-4.0.0-alpha-1	N/A	N/A	N/A	☆
⊗	☁	branch-4.0.0-alpha-2	N/A	2 yr 9 mo #2	4 hr 25 min	☆
⊗	☁	branch-4.0.0-beta-1	N/A	1 yr 11 mo #1	6 hr 31 min	☆
ⓘ	☀	branch-4.1	2 days 7 hr #23	N/A	3 hr 15 min	☆
⊗	☁	dependabot/maven/itests/qtest-druid/org.apache.kafka-kafka_2.12-3.7.2	N/A	18 days #1	50 min	☆
ⓘ	☀	dependabot/maven/service/com.nimbusds-nimbus-jose-jwt-10.0.2	9 days 21 hr #1	N/A	8 hr 10 min	☆
✅	☀	dependabot/maven/standalone-metastore/org.apache.commons-commons-lang3-3.18.0	9 days 12 hr #1	N/A	5 hr 14 min	☆
✅	☀	HiveAggregateReduceFunctionsRule	14 days #1	N/A	4 hr 47 min	☆
⊗	☀	master	5 days 5 hr #2615	12 days #2603	5 hr 44 min	☆



Jenkins - Precommit job

Configured via:

<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

Task decomposition:

1. Checkout
2. Prechecks
3. Compile
4. Upload
5. Parallel checks:
 - Metastore initialization
 - Derby
 - Postgres
 - Javadoc
 - Nightly build
 - Sonar
 - Tests (22 Splits)



Jenkins - Precommit job

Configured via:

<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

Task decomposition:

1. Checkout (~1m)
2. Prechecks (~1m)
3. Compile (~40m)
4. Upload (~10m)
5. Parallel checks:
 - Metastore initialization (~5m)
 - Derby
 - Postgres
 - Javadoc (~1hr)
 - Nightly build (~1hr)
 - Sonar (~30min)
 - Tests in #22 Splits (~1.5hrs)

Jenkins - Precommit job



 hive-precommit 1

PipelineChangesTestsArtifacts

Pull Request: PR-5989

1h 17m 41s

No changes

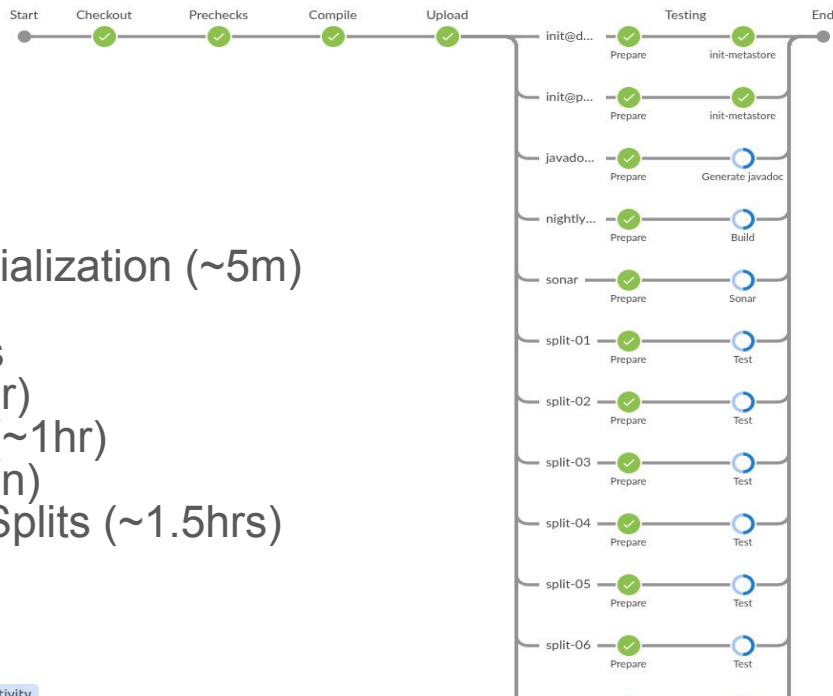
Commit: 59ec123

-

Pull request #5989 opened

Task decomposition:

1. Checkout (~1m)
2. Prechecks (~1m)
3. Compile (~40m)
4. Upload (~10m)
5. Parallel checks:
 - Metastore initialization (~5m)
 - Derby
 - Postgres
 - Javadoc (~1hr)
 - Nightly build (~1hr)
 - Sonar (~30min)
 - Tests in #22 Splits (~1.5hrs)





Jenkins - Debugging

- Inspect full Jenkins logs (consoleText):
<https://ci.hive.apache.org/job/hive-precommit/job/master/2618/consoleText>
- Download and check test-results.tgz:
- <https://ci.hive.apache.org/job/hive-precommit/view/change-requests/job/PR-5976/lastSuccessfulBuild/artifact/test-results.tgz>
 - Maven Surefire reports
 - Hive unit test logs
- Check pipeline steps to get info about underlying resources (K8S):
<https://ci.hive.apache.org/job/hive-precommit/view/change-requests/job/PR-5976/lastCompletedBuild/flowGraphTable/>
- Reproduce split failures using **inclusions.txt** info



Jenkins - Image & Deployment

Image: <https://github.com/apache/hive/pull/5331>

Registry: <https://hub.docker.com/r/apache/hive-ci-jenkins/>

Deployment: `kubectl get deployment jenkins -o yaml`

Old deployment:

<https://github.com/kgyrtkirk/hive-test-kube/blob/master/k8s/jenkins-deployment.yaml>



Jenkins - Artifactory integration

Configuration via Jenkinsfile + settings.xml

(<https://ci.hive.apache.org/manage/configfiles/>)

```
<?xml version="1.0" encoding="UTF-8"?>
<settings xsi:schemaLocation="http://maven.apache.org/SETTINGS/1.1.0
http://maven.apache.org/xsd/settings-1.1.0.xsd" xmlns="http://maven.apache.org/SETTINGS/1.1.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <mirrors>
    <mirror>
      <mirrorOf>*</mirrorOf>
      <name>wonder</name>
      <url>http://artifactory/artifactory/wonder</url>
      <id>wonder</id>
    </mirror>
  </mirrors>
</settings>
```



JFrog Artifactory

<http://35.239.52.74/artifactory/webapp/#/home>

The screenshot displays the JFrog Artifactory web application interface. At the top, a green header bar contains the JFrog logo, the text "JFrog Artifactory", a search icon, and links for "Help" and "Log In". Below the header, a dark sidebar on the left lists various navigation icons. The main content area features a large heading "Artifactory is happily serving 18,609 artifacts" with the version "Artifactory Version 6.18.1" underneath. This is followed by a "Quick Search" section with tabs for "Package Search", "Archive Search", "Property Search", "Checksum Search", and "JCenter Search". Below the search section is a grid of six icons: "User Guide", "Webinar Signup", "Support Portal", "Stack Overflow", "Blog", and "Rest API". To the right of the search section is a "Set Me Up" panel with a "Filter by Repository Key" dropdown and a list of repositories including "wonder", "artifactory-build-info", "central", "clojars", "confluent", "shibboleath", "spring-plugins", "spring-repository", and "typesafe". Further right is a "Last Deployed Builds" panel showing "No builds to display." and a link to "Learn how to integrate your build information with Artifactory." Below this is a "Most Downloaded Artifacts" panel listing several artifacts with their sizes and names, such as "central-cache/org/easystesting/fest-assert/1.4/fest-assert-1.4.pom". At the bottom, a horizontal bar contains icons and labels for "High Availability", "JFrog Bintray Distribution", "JFrog Xray", "JFrog CLI", "Build Integration", "Docker", "Repository Replication", "Multipush Replication", and "AQL".

Artifactory is happily serving 18,609 artifacts
Artifactory Version 6.18.1

Quick Search
Or go to >

Package Search | Archive Search | Property Search | Checksum Search | JCenter Search

User Guide | Webinar Signup | Support Portal | Stack Overflow | Blog | Rest API

Set Me Up
Filter by Repository Key

- wonder
- artifactory-build-info
- central
- clojars
- confluent
- shibboleath
- spring-plugins
- spring-repository
- typesafe

Last Deployed Builds (16/07/25 12:17:55)

No builds to display.
Learn how to integrate your build information with Artifactory.

Most Downloaded Artifacts (16/07/25 12:17:55)

- 342.1 K central-cache/org/easystesting/fest-assert/1.4/fest-assert-1.4.pom
- 342.1 K central-cache/org/easystesting/fest/1.0.8/fest-1.0.8.pom
- 342.0 K central-cache/org/easystesting/fest-util/1.1.6/fest-util-1.1.6.pom
- 342.0 K central-cache/junit/junit/4.0/junit-4.0.pom
- 341.9 K central-cache/org/apache/maven/surefire/surefire-providers/3.0.0-M4/sur...

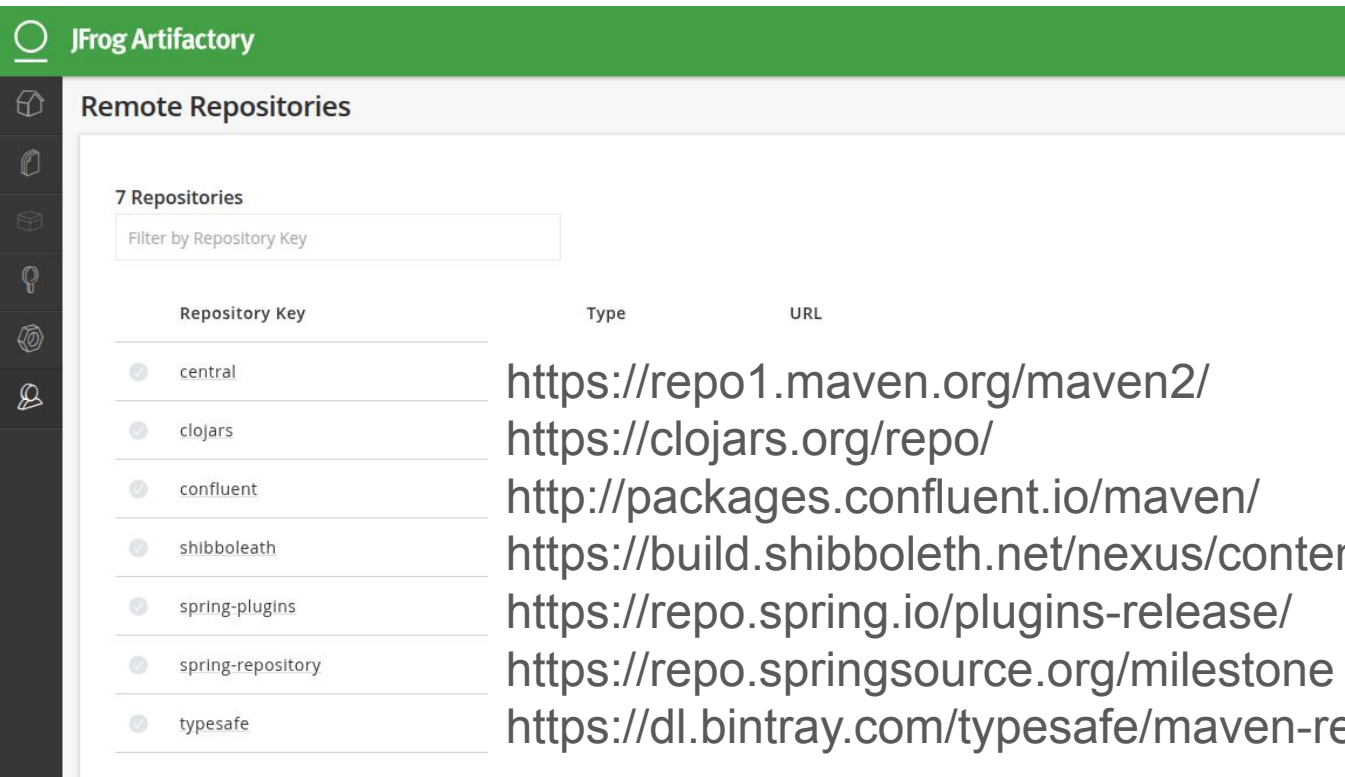
High Availability | JFrog Bintray Distribution | JFrog Xray | JFrog CLI | Build Integration | Docker | Repository Replication | Multipush Replication | AQL

JFrog Artifactory - Access & Permissions

- Limited access
 - No LDAP
 - No SSO
 - No Auth plugins
- Privileges/Access can be granted to Hive committers/PMC per request

JFrog Artifactory - Remote Repositories

<http://35.239.52.74/artifactory/webapp/#/home>



The screenshot shows the JFrog Artifactory web interface. At the top is a green header with the JFrog Artifactory logo and name. Below this is a sidebar with navigation icons. The main content area is titled 'Remote Repositories' and shows a list of 7 repositories. A search bar labeled 'Filter by Repository Key' is present. The repositories are listed in a table with columns for Repository Key, Type, and URL.

Repository Key	Type	URL
central		https://repo1.maven.org/maven2/
clojars		https://clojars.org/repo/
confluent		http://packages.confluent.io/maven/
shibboleth		https://build.shibboleth.net/nexus/content/repositories/releases/
spring-plugins		https://repo.spring.io/plugins-release/
spring-repository		https://repo.springsource.org/milestone
typesafe		https://dl.bintray.com/typesafe/maven-releases/

JFrog Artifactory - Virtual Repositories

<http://35.239.52.74/artifactory/webapp/#/home>

 JFrog Artifactory



Virtual Repositories

1 Repository

Repository Key	Type	Included Repositories	Selected Repositories
☒ wonder	<i>m</i> Maven	7	7 central, confluent, spring-repository, typesafe, clojars, spring-plugins, shibboleath

JFrog Artifactory - Image & Deployment

Image: `docker.bintray.io/jfrog/artifactory-oss:6.18.1`

Deployment: `kubectl get deployment artifactory -o yaml`

Old deployment:

<https://github.com/kgyrtekirk/hive-test-kube/blob/master/k8s/artifactory.yaml>

rsync

RSYNC



Controlled via Jenkinsfile

- saveWS: `rsync -rltDq --stats archive.tar rsync://rsync/data/$LOCKED_RESOURCE`
- loadWS: `rsync -rltDq --stats rsync://rsync/data/$LOCKED_RESOURCE archive.tar`

RSYNC



Controlled via Jenkinsfile

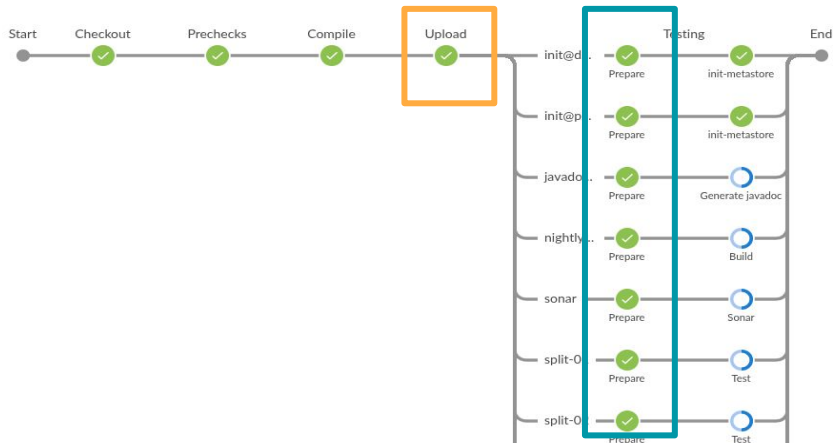
- **saveWS**: `rsync -rltDq --stats archive.tar rsync://rsync/data/$LOCKED_RESOURCE`
- **loadWS**: `rsync -rltDq --stats rsync://rsync/data/$LOCKED_RESOURCE archive.tar`

hive-precommit 1

Pipeline Changes Tests Artifacts Logout

Pull Request: PR-5989 1h 17m 41s No changes

Commit: 59ec123 Pull request #5989 opened



RSYNC - Image & Deployment



Image: <https://github.com/kgyrtkirk/hive-test-kube/blob/master/htk-rsync/Dockerfile>

Registry: <https://hub.docker.com/r/kgyrtkirk/htk-rsync>

Deployment: `kubectl get deployment rsync -o yaml`

Old Deployment:

<https://github.com/kgyrtkirk/hive-test-kube/blob/master/k8s/rsync.yaml>



Google Cloud



<https://console.cloud.google.com/home/dashboard?project=gcp-hive-upstream>

The screenshot shows the Google Cloud Platform console dashboard for the project 'gcp-hive-upstream'. The interface includes a left-hand navigation menu with sections for 'Google Cloud' (Cloud Hub, Cloud overview, Solutions, Recently visited), 'PINNED PRODUCTS' (IAM & Admin, Billing, APIs & Services, Marketplace, Compute Engine, Cloud Storage, VPC Network, Kubernetes Engine, BigQuery, SQL, Cloud Run), and 'PRODUCTS' (Explore and enable APIs, Deploy a prebuilt solution, Add dynamic logging to a running application). The main dashboard area is divided into several widgets:

- Project info:** Displays project details such as Project name (gcp-hive-upstream), Project number (956759479571), and Project ID (gcp-hive-upstream). It includes a link to 'Go to project settings'.
- Compute Engine:** Shows a line graph for CPU (%) utilization over time. The current instance/cpu utilization is 59.59%. A note states: 'The data might not be complete because some regions are down.' A link to 'Go to Compute Engine' is provided.
- Resources:** Lists various Google Cloud services available to the project, including BigQuery, SQL, Compute Engine, Storage, Cloud Run functions, and Cloud Run.
- Getting Started:** Provides links to 'Explore and enable APIs', 'Deploy a prebuilt solution', and 'Add dynamic logging to a running application'.
- Google Cloud Platform status:** Indicates 'All services normal' and provides a link to 'Go to Cloud status dashboard'.
- Billing:** Offers a 'Take a tour of billing' and a link to 'View detailed charges'.
- Monitoring:** Includes options to 'Create my dashboard', 'Set up alerting policies', 'Create uptime checks', and a link to 'Go to Monitoring'.
- API Error Reporting:** Shows 'Top errors in last 24 hours', including a specific error: 'pod didn't trigger scale-up: 1 max node group size reached' with 1,122 occurrences, and 'KubernetesClientTimeoutException BaseOperation.waitUntilCondition()' with 166 occurrences.

The top of the dashboard features a search bar and a 'CUSTOMIZE' link.

Google Cloud - Sponsor



<https://console.cloud.google.com/home/dashboard?project=gcp-hive-upstream>

The screenshot displays the Google Cloud console dashboard for the project 'gcp-hive-upstream'. The interface includes a left sidebar with navigation links for Cloud Hub, Cloud overview, Solutions, Recently visited, and Pinned Products (IAM & Admin, Billing, APIs & Services, Marketplace, Compute Engine, Cloud Storage, VPC Network, Kubernetes Engine, BigQuery, SQL, Cloud Run, Google Maps Plat...). The main content area is divided into several sections:

- Project info:** Displays project details such as Project name (gcp-hive-upstream), Project number (956759479571), and Project ID (gcp-hive-upstream). It also includes a link to 'Go to project settings'.
- Compute Engine:** Shows a line graph for CPU (%) usage over time, with a peak around 11:45 AM. Below the graph, it indicates 'Instance/cpu utilization: 59.59%' and provides a link to 'Go to Compute Engine'.
- Resources:** Lists various resources including BigQuery, SQL, Compute Engine, Storage, Cloud Run functions, and Cloud Run.
- Getting Started:** Provides links to 'Explore and enable APIs', 'Deploy a prebuilt solution', and 'Add dynamic logging to a running application'.
- Google Cloud Platform status:** Indicates 'All services normal' and includes a link to 'Go to Cloud status dashboard'.
- Billing:** Includes a link to 'Take a tour of billing' and 'View detailed charges'.
- Monitoring:** Offers options to 'Create my dashboard', 'Set up alerting policies', 'Create uptime checks', and 'View all dashboards'.
- RPI Error Reporting:** Shows 'Top errors in last 24 hours' with a table listing errors like 'pod didn't trigger scale-up: 1 max node group size reached' and 'KubernetesClientTimeoutException'.

CLOUDEIRA

Google Cloud - Access & Permissions



<https://console.cloud.google.com/home/dashboard?project=gcp-hive-upstream>

The screenshot shows the Google Cloud console interface. The top navigation bar includes the Google Cloud logo, the project name 'gcp-hive-upstream', a search bar, and user profile information. The left sidebar lists various services, with 'IAM & Admin' highlighted by an orange rectangle. The main dashboard area contains the following list:

- Access restricted to Cloudera employees
- Permission/privileges granted upon request
- Report problem/Suggest improvement:
 - Send email to dev@hive.apache.org
 - Submit a [Hive JIRA ticket](#)

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

hive-test-kube

```
$ gcloud container node-pools list --cluster=hive-test-kube --zone=us-central1-c
```

NAME	MACHINE_TYPE	DISK_SIZE_GB	NODE_VERSION
slave-pool9	e2-custom-12-65536	160	1.31.8-gke.1045000
core-pool2	custom-2-13312	100	1.31.8-gke.1045000

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

hive-test-kube

```
$ gcloud container node-pools describe core-pool2 --cluster=hive-test-kube  
--zone=us-central1-c  
autoscaling: {}  
config:  
  diskSizeGb: 100  
  diskType: pd-ssd
```

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

hive-test-kube

```
$ gcloud container node-pools describe slave-pool9 --cluster=hive-test-kube
```

```
--zone=us-central1-c
```

```
autoscaling:
```

```
  enabled: true
```

```
  locationPolicy: BALANCED
```

```
  maxNodeCount: 8
```

```
config:
```

```
  diskSizeGb: 160
```

```
  diskType: pd-ssd
```

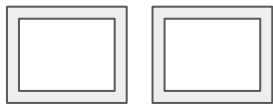
Google Cloud - Google Kubernetes Engine (GKE)



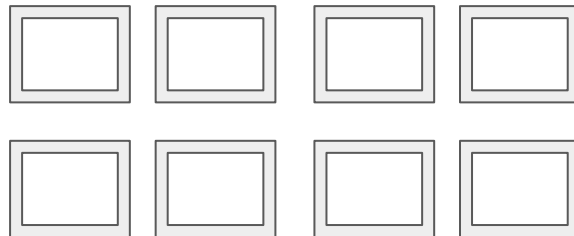
<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

hive-test-kube

core-pool2
2 nodes



slave-pool9
0-8 nodes



Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

hive-test-kube

core-pool2
2 nodes

- Machine type: custom-2-13312
- Boot disk: 100GB SSD

slave-pool9
0-8 nodes

- Machine type: e2-custom-12-65536
- Boot disk: 160GB SSD

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

```
$ gcloud compute machine-types describe custom-2-13312 --zone=us-central1-c
```

```
description: Custom created machine type.
```

```
guestCpus: 2
```

```
id: '867583407104'
```

```
isSharedCpu: false
```

```
kind: compute#machineType
```

```
maximumPersistentDisks: 128
```

```
maximumPersistentDisksSizeGb: '263168'
```

```
memoryMb: 13312
```

```
name: custom-2-13312
```

```
selfLink:
```

```
https://www.googleapis.com/compute/v1/projects/gcp-hive-upstream/zones/us-central1-c/machineTypes/custom-2-13312
```

```
zone: us-central1-c
```

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

```
$ gcloud compute machine-types describe e2-custom-12-65536 --zone=us-central1-c
```

```
description: Custom created machine type.
```

```
guestCpus: 12
```

```
id: '5205500428288'
```

```
isSharedCpu: false
```

```
kind: compute#machineType
```

```
maximumPersistentDisks: 128
```

```
maximumPersistentDisksSizeGb: '524288'
```

```
memoryMb: 65536
```

```
name: e2-custom-12-65536
```

```
selfLink:
```

```
https://www.googleapis.com/compute/v1/projects/gcp-hive-upstream/zones/us-central1-c/machineTypes/e2-custom-12-65536
```

```
zone: us-central1-c
```

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

hive-test-kube

core-pool2 2 nodes

- Machine type: custom-2-13312
 - 2 CPUs
 - 13312 Mb memory
- Boot disk: 100GB SSD

slave-pool9 0-8 nodes

- Machine type: e2-custom-12-65536
 - 12 CPUs
 - 65536 Mb memory
- Boot disk: 160GB SSD

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

\$ kubectl get nodes

NAME	STATUS	ROLES	AGE	VERSION
gke-hive-test-kube-core-pool2-41eb7daa-ob7l	Ready	<none>	9h	v1.31.8-gke.1113000
gke-hive-test-kube-core-pool2-41eb7daa-xoic	Ready	<none>	9h	v1.31.8-gke.1113000
gke-hive-test-kube-slave-pool9-bbcf4d84-5w9v	Ready	<none>	5h	v1.31.8-gke.1113000
gke-hive-test-kube-slave-pool9-bbcf4d84-b4l8	Ready	<none>	5h1m	v1.31.8-gke.1113000
gke-hive-test-kube-slave-pool9-bbcf4d84-k5s5	Ready	<none>	5h	v1.31.8-gke.1113000
gke-hive-test-kube-slave-pool9-bbcf4d84-pvdh	Ready	<none>	3h41m	v1.31.8-gke.1113000
gke-hive-test-kube-slave-pool9-bbcf4d84-t96j	Ready	<none>	5h40m	v1.31.8-gke.1113000
gke-hive-test-kube-slave-pool9-bbcf4d84-w2vm	Ready	<none>	5h	v1.31.8-gke.1113000
gke-hive-test-kube-slave-pool9-bbcf4d84-wgpm	Ready	<none>	5h1m	v1.31.8-gke.1113000

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

hive-test-kube

core-pool2
2 nodes



JFrog
ARTIFACTORY



slave-pool9
0-8 nodes



hive-precommit-XX pods

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

\$ kubectl get pods

NAME	READY	STATUS	RESTARTS	AGE
artifactory-65b4c87666-6mj9h	1/1	Running	0	30d
hive-precommit-master-2615-mr7m2-v5fp8-6lg51	3/3	Running	0	12s
hive-precommit-pr-5973-6-cw3lk-3jb6c-0hb1p	3/3	Running	0	34m
hive-precommit-pr-5977-4-bl12t-6hjdc-94fz4	3/3	Running	0	14m
jenkins-558bfdb466-89tm9	1/1	Running	0	30d
rsync-7874446954-cpw5k	1/1	Running	0	30d
web-6fff6cdf7d-8bzsb	1/1	Running	0	30d

Google Cloud - Google Kubernetes Engine (GKE)



<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

```
$ kubectl get pods
```

```
NAME
```

```
artifactory-65b4c87666-6mj9h
```

```
hive-precommit-master-2615-mr7
```

```
hive-precommit-pr-5973-6-cw3lk-
```

```
hive-precommit-pr-5977-4-bl12t-6
```

```
jenkins-558bfdb466-89tm9
```

```
rsync-7874446954-cpw5k
```

```
web-6fff6cdf7d-8bzsb
```

core-pool2

Google Cloud - Google Kubernetes Engine (GKE)



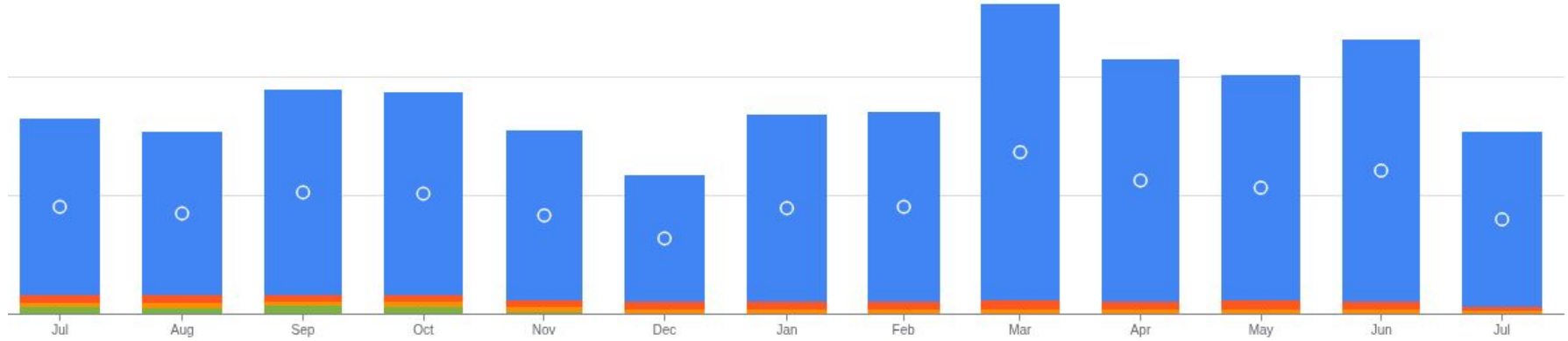
<https://console.cloud.google.com/kubernetes/clusters/details/us-central1-c/hive-test-kube/details?project=gcp-hive-upstream>

```
$ kubectl get pods
```

NAME	READY	STATUS
artifactory-65b4c87666-6mj9h	1/1	Running
hive-precommit-master-2615-mr7m2-v5fp8-6lg51	1/1	Running
hive-precommit-pr-5973-6-cw3lk-3jb6c-0hb1p	3/3	Running
hive-precommit-pr-5977-4-bl12t-6hjdk-94fz4	3/3	Running
jenkins-558bfdb466-89tm9	1/1	Running
rsync-7874446954-cpw5k	1/1	Running
web-6fff6cdf7d-8bzsb	1/1	Running

slave-pool9

Google Cloud - Cost



Forecasted cost ?

[Download CSV](#)

- Thousands of dollars per month
- Proportional to the number of CI runs

Service

- Compute Engine
- Kubernetes Engine
- ◆ Networking
- ▼ Support
- ▲ Cloud Logging
- Cloud Monitoring



Hive Precommit Pods - Naming

<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

hive-precommit-pr-5973-6-0hz7r-tx6tr-379qt

Branch

Run

Generated ID

hive-precommit-master-2615-mr7m2-v5fp8-6lg51

Hive Precommit Pods - Containers



<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

```
$ kubectl get pods hive-precommit-pr-5973-6-z037r-plrf7-w1q7m -o  
jsonpath='{.spec.containers[*].name}'  
dind hdb jnlp
```



Hive Precommit Pods - Containers

<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

containers: [

```
  containerTemplate(name: 'hdb', image: 'ayushtkn/hive-dev-box:executor', ttyEnabled: true, command: 'tini -- cat',
    alwaysPullImage: true,
    resourceRequestCpu: '1800m',
    resourceLimitCpu: '8000m',
    resourceRequestMemory: '6400Mi',
    resourceLimitMemory: '12000Mi',
    resourceRequestEphemeralStorage: '15Gi',
    resourceLimitEphemeralStorage: '30Gi',
    envVars: [...]),
  containerTemplate(name: 'dind', image: 'docker:20.10-dind',
    alwaysPullImage: true,
    privileged: true,
    envVars: [...]),]
```



Hive Precommit Pods - Scheduling

<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

containers: [

```
  containerTemplate(name: 'hdb', image: 'ayushtkn/hive-dev-box:executor', ttyEnabled: true, command: 'tini -- cat',
    alwaysPullImage: true,
    resourceRequestCpu: '1800m',
    resourceLimitCpu: '8000m',
    resourceRequestMemory: '6400Mi',
    resourceLimitMemory: '12000Mi',
    resourceRequestEphemeralStorage: '15Gi',
    resourceLimitEphemeralStorage: '30Gi',
    envVars: [...]),
  containerTemplate(name: 'dind', image: 'docker:20.10-dind',
    alwaysPullImage: true,
    privileged: true,
    envVars: [...]),]
```

slave-pool9
0-8 nodes

- Machine type: e2-custom-12-65536
 - 12 CPUs
 - 65536 Mb memory
- Boot disk: 160GB SSD



Hive Precommit Pods - Scheduling

<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

slave-pool9

CPU: $12 / 1.8 = 6.6$

Memory (Mi): $65536 / 6400 = 10.24$

Disk (Gi): $160 / 15 = 10.6$

Each node can run at most ~6 pods concurrently

Node pool capacity: ~48 concurrent pods



Hive Precommit Pods - Scheduling

<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

slave-pool9

(gke-hive-test-kube-slave-pool9-bbcf4d84-t96)

```
$ kubectl get pods --all-namespaces --field-selector spec.nodeName=gke-hive-test-kube-slave-pool9-bbcf4d84-t96j
```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
default	hive-precommit-pr-5989-1-15sdd-hj6r7-w8mb8	3/3	Running	0	20m
default	hive-precommit-pr-5989-1-1r9m4-1j475-2r53k	3/3	Running	0	20m
default	hive-precommit-pr-5989-1-6kxk6-hhf94-61723	3/3	Running	0	20m
default	hive-precommit-pr-5989-1-hk9x9-gq7v2-q6744	3/3	Running	0	20m
default	hive-precommit-pr-5989-1-p43nb-d6kmm-grdvv	3/3	Running	0	20m
kube-system	fluentbit-gke-2d6kq	3/3	Running	0	6h8m
kube-system	gke-metrics-agent-p46qs	3/3	Running	0	6h8m
kube-system	kube-proxy-gke-hive-test-kube-slave-pool9-bbcf4d84-t96j	1/1	Running	0	6h8m
kube-system	pdcsi-node-69p99	2/2	Running	0	6h8m



Hive Precommit Pods - Scheduling

hive-precommit 1

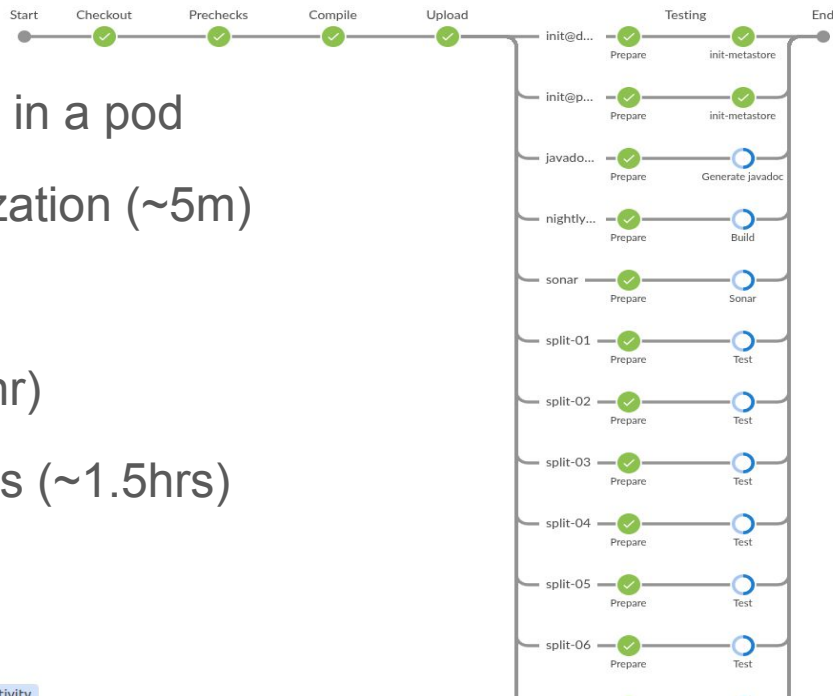
PipelineChangesTestsArtifacts

Pull Request: PR-5989 ☒ 1h 17m 41s No changes
Commit: 59ec123 - Pull request #5989 opened

Each task/action runs in a pod

27 parallel tasks:

- Metastore initialization (~5m)
 - Derby
 - Postgres
- Javadoc (~1hr)
- Nightly build (~1hr)
- Sonar (~30min)
- Tests in #22 Splits (~1.5hrs)





Hive Precommit Pods - Scheduling

<https://github.com/apache/hive/blob/5512ffda602bef605ad144f9040676149cc2c960/Jenkinsfile>

slave-pool9

CPU: $12 / 1.8 = 6.6$

Memory (Mi): $65536 / 6400 = 10.24$

Disk (Gi): $160 / 15 = 10.6$

Each node can run at most ~6 pods concurrently

Node pool capacity: $8 * 6 = \sim 48$ concurrent pods

CI Run capacity: $48 / 27 = \sim 2$ full runs

Thank you

